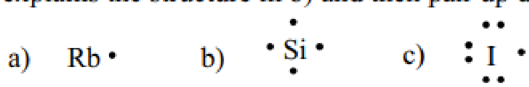
9.1 a) Larger ionization energy decreases metallic character.
b) Larger atomic radius increases metallic character.
c) Larger number of outer electrons decreases metallic character.
d) Larger effective nuclear charge decreases metallic character.

9.4 Metallic behavior increases to the left and down on the periodic table.
a) Cs is more metallic since it is further down the alkali metal group than Na.
b) Rb is more metallic since it is both to the left and down from Mg.
c) As is more metallic since it is further down Group 5A than N.

9.6 Ionic bonding occurs between metals and nonmetals, covalent bonding between nonmetals, and metallic bonds between metals.
a) Bond in CsF is ionic because Cs is a metal and F is a nonmetal.
b) Bonding in N_2 is covalent because N is a nonmetal.
c) Bonding in Na(*s*) is metallic because this is a monatomic, metal solid.

9.8 a) Bonding in O_3 would be covalent since O is a nonmetal.
b) Bonding in $MgCl_2$ would be ionic since Mg is a metal and Cl is a nonmetal.
c) Bonding in BrO_2 would be covalent since both Br and O are nonmetals.

9.10 Lewis electron-dot symbols show valence electrons as dots. Place one dot at a time on the four sides (this method explains the structure in b) and then pair up dots until all valence electrons are used.

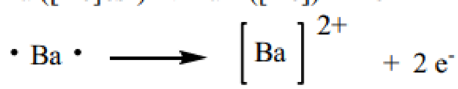


9.14 a) Assuming X is an A-group element, the number of dots (valence electrons) equals the group number. Therefore, X is a 6A(16) element. Its general electron configuration is [noble gas]ns^2np^4, where *n* is the energy level.
b) X has three valence electrons and is a 3A(13) element with general e^- configuration [noble gas]ns^2np^1.

9.17 Because the lattice energy is the result of electrostatic attractions among the oppositely charged ions, its magnitude depends on several factors, including ionic size, ionic charge, and the arrangement of ions in the solid. For a particular arrangement of ions, the lattice energy increases as the charges on the ions increase and as their radii decrease.

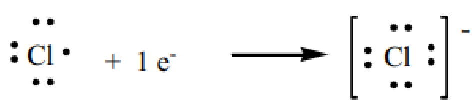
9.20 a) Barium is a metal and loses 2 electrons to achieve a noble gas configuration:

$Ba\ ([Xe]6s^2) \rightarrow Ba^{2+}\ ([Xe]) + 2\ e^-$

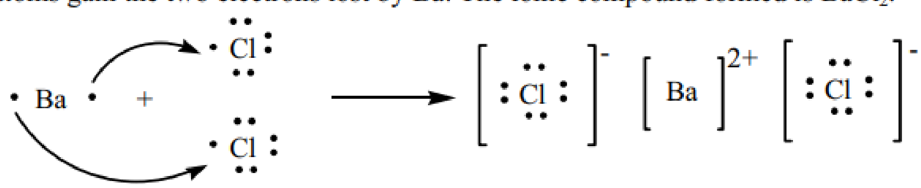


Chlorine is a nonmetal and gains 1 electron to achieve a noble gas configuration:

$Cl\ ([Ne]3s^23p^5) + 1\ e^- \rightarrow Cl^-\ ([Ne]3s^23p^6)$

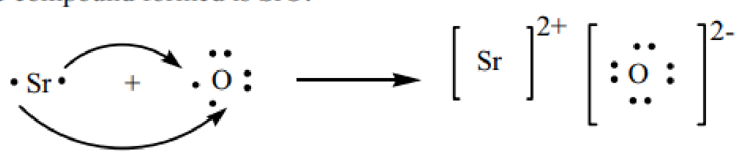


Two Cl atoms gain the two electrons lost by Ba. The ionic compound formed is $BaCl_2$.

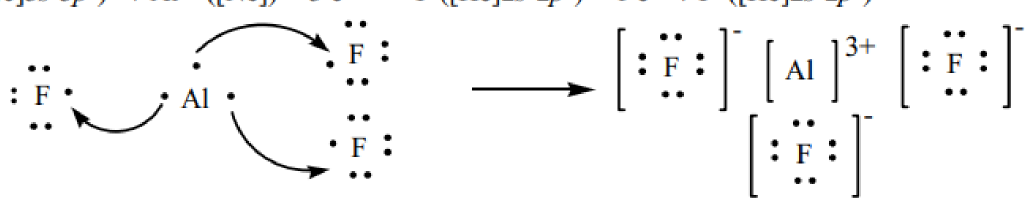


b) $Sr\ ([Kr]5s^2) \rightarrow Sr^{2+}\ ([Kr]) + 2\ e^-$ $\quad$ $O\ ([He]2s^22p^4) + 2\ e^- \rightarrow O^{2-}\ ([He]2s^22p^6)$

The ionic compound formed is SrO.

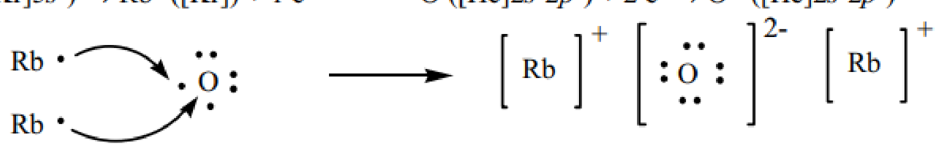


c) $Al\ ([Ne]3s^23p^1) \rightarrow Al^{3+}\ ([Ne]) + 3\ e^-$ $\quad$ $F\ ([He]2s^22p^5) + 1\ e^- \rightarrow F^-\ ([He]2s^22p^6)$



The ionic compound formed is AlF_3.

d) $Rb\ ([Kr]5s^1) \rightarrow Rb^+\ ([Kr]) + 1\ e^-$ $\quad$ $O\ ([He]2s^22p^4) + 2\ e^- \rightarrow O^{2-}\ ([He]2s^22p^6)$



The ionic compound formed is Rb_2O.

9.22 a) X in XF_2 is a cation with +2 charge since the anion is F^- and there are two fluoride ions in the compound. Group 2A(2) metals form +2 ions.
b) X in MgX is an anion with -2 charge since Mg^{2+} is the cation. Elements in Group 6A(16) form -2 ions.
c) X in X_2SO_4 must be a cation with +1 charge since the polyatomic sulfate ion has a charge of -2. X comes from Group 1A(1).

9.24 a) X in X_2O_3 is a cation with +3 charge. The oxygen in this compound has a -2 charge. To produce an electrically neutral compound, 2 cations with +3 charge bond with 3 anions with -2 charge: 2(+3) + 3(-2) = 0. Elements in Group 3A(13) form +3 ions.
b) The carbonate ion, CO_3^{2-}, has a -2 charge, so X has a +2 charge. Group 2A(2) elements form +2 ions.
c) X in Na_2X has a -2 charge, balanced with the +2 overall charge from the two Na atoms. Group 6A(16) elements gain 2 electrons to form -2 ions with a noble gas configuration.

9.26 a) BaS would have the higher lattice energy since the charge on each ion is twice the charge on the ions in CsCl and lattice energy is greater when ionic charges are larger.
b) LiCl would have the higher lattice energy since the ionic radius of Li^+ is smaller than that of Cs^+ and lattice energy is greater when the distance between ions is smaller.

9.28 a) BaS has the lower lattice energy because the ionic radius of Ba^{2+} is larger than Ca^{2+}. A larger ionic radius results in a greater distance between ions. The lattice energy decreases with increasing distance between ions.
b) NaF has the lower lattice energy since the charge on each ion (+1, -1) is half the charge on the Mg^{2+} and O^{2-} ions.

9.30 Lattice energy can be found from the fact that the sum of the energies from each step which equals the heat of formation for solid sodium chloride.

$$\Delta H_f^\circ + \Delta H_{Na(s)\rightarrow Na(g)} + \frac{1}{2}\Delta H_{Cl_2(g)\rightarrow 2Cl(g)} + \Delta H_{Na(g)\rightarrow Na^+(g)+e^-} + \Delta H_{Cl(g)+e^-\rightarrow Cl^-(g)} = \Delta H_{lattice}$$

$\Delta H_{lattice}$ = -41 kJ + 109 kJ + (1/2) (243 kJ) + 496 kJ + (-349 kJ) = 336.5 = 336 kJ

The lattice energy for NaCl is less than that for LiF, which is expected since lithium and fluoride ions are smaller than sodium and chloride ions.

9.33 An analogous Born-Haber cycle has been described in Figure 9.6 for LiF. Use Figure 9.6 as a basis for this diagram and solve for step 4, the electron affinity of F.

1) $K(s) \rightarrow K(g)$	$\Delta H_1 = 90$ kJ
2) $K(g) \rightarrow K^+(g) + e^-$	$\Delta H_2 = 419$ kJ
3) $1/2\ F_2(g) \rightarrow F(g)$	$\Delta H_3 = (1/2)\ (159)$ kJ
4) $F(g) + e^- \rightarrow F^-(g)$	$\Delta H_4 = ? = EA$
5) $K^+ + F^-(g) \rightarrow KF(s)$	$\Delta H_5 = -821$ kJ
6) $K(s) + 1/2\ F_2(g) \rightarrow KF(s)$	$\Delta H_f = -569$ kJ

$$\Delta H_4 = EA = \Delta H_f - (\Delta H_1 + \Delta H_2 + \Delta H_3 + \Delta H_5)$$

$$EA = -569 - (90 + 419 + \frac{159}{2} + (-821))$$

$$EA = -336 \text{ kJ}$$

(Fig 9.6 is only for the fourth edition of Silberberg. Find the analogous figure in your edition.)

9.35 The bond energy is the energy required to overcome the attraction between H atoms and Cl atoms in one mole of HCl molecules in the gaseous state. Energy input is needed to break bonds, so bond energy is always endothermic and $\Delta H^\circ_{\text{bond breaking}}$ is positive. The same amount of energy needed to break the bond is released upon its formation, so $\Delta H^\circ_{\text{bond forming}}$ has the same magnitude as $\Delta H^\circ_{\text{bond breaking}}$, but opposite in sign (always exothermic and negative).

9.39 a) I-I < Br-Br < Cl-Cl. Bond strength increases as the atomic radii of atoms in the bond decreases. Atomic radii decrease up a group in the periodic table, so I is the largest and Cl is the smallest of the three.
b) S-Br < S-Cl < S-H. Radius of H is the smallest and Br is the largest, so the bond strength for S-H is the greatest and that for S-Br is the weakest.
c) C-N < C=N < C≡N. Bond strength increases as the number of electrons in the bond increases. The triple bond is the strongest and the single bond is the weakest.

9.44 Reaction between molecules requires the breaking of existing bonds and the formation of new bonds. Substances with weak bonds are more reactive than are those with strong bonds because less energy is required to break weak bonds.

9.46 For methane: $CH_4(g) + 2\ O_2(g) \rightarrow CO_2(g) + 2\ H_2O(l)$ which requires that 4 C-H bonds and 2 O=O bonds be broken and 2 C=O bonds and 4 O-H bonds be formed.
For formaldehyde: $CH_2O(g) + O_2(g) \rightarrow CO_2(g) + H_2O(l)$ which requires that 2 C-H bonds, 1 C=O bond and 1 O=O bond be broken and 2 C=O bonds and 2 O-H bonds be formed.
The fact that methane contains more C-H bonds and fewer C-O bonds than formaldehyde suggests that more energy is released in the combustion of methane than of formaldehyde.

9.52 Electronegativity increases from left to right across a period (except for the noble gases) and increases from bottom to top within a group. Fluorine (F) and oxygen (O) are the two most electronegative elements. Cesium (Cs) and francium (Fr) are the two least electronegative elements.

9.54 The H-O bond in water is polar covalent, because the two atoms differ in electronegativity which results in an unequal sharing of the bonding electrons. A nonpolar covalent bond occurs between two atoms with identical electronegativities where the sharing of bonding electrons is equal. Although electron sharing occurs to a very small extent in some ionic bonds, the primary force in ionic bonds is attraction of opposite charges resulting from electron transfer between the atoms.

9.57 a) Si < S < O, sulfur is more electronegative than silicon since it is located further to the right on the table. Oxygen is more electronegative than sulfur since it is located nearer the top of the table.
b) Mg < As < P, magnesium is the least electronegative because it lies on the left side of the periodic table and phosphorus and arsenic on the right side. Phosphorus is more electronegative than arsenic because it is higher on the table.

9.59 Electronegativity generally increases up a group and left to right across a period.
a) N > P > Si, nitrogen is above P in Group 5(A)15 and P is to the right of Si in period 3.
b) As > Ga > Ca, all three elements are in Period 4, with As the right-most element.

9.65 a) Bonds in S_8 are nonpolar covalent. All the atoms are nonmetals so the substance is covalent and bonds are nonpolar because all the atoms are of the same element.
b) Bonds in RbCl are ionic because Rb is a metal and Cl is a nonmetal.
c) Bonds in PF_3 are polar covalent. All the atoms are nonmetals so the substance is covalent. The bonds between P and F are polar because their electronegativity differs. (By 1.9 units for P-F)
d) Bonds in SCl_2 are polar covalent. S and Cl are nonmetals and differ in electronegativity.
(By 0.5 unit for S-Cl)
e) Bonds in F_2 are nonpolar covalent. F is a nonmetal. Bonds between two atoms of the same element are nonpolar.

f) Bonds in SF_2 are polar covalent. S and F are nonmetals that differ in electronegativity. (By 1.5 units for S-F)
Increasing bond polarity: $SCl_2 < SF_2 < PF_3$

9.67 Increasing ionic character occurs with increasing ΔEN.
a) $\Delta EN_{HBr} = 0.7$, $\Delta EN_{HCl} = 0.9$, $\Delta EN_{HI} = 0.4$
b) $\Delta EN_{HO} = 1.4$, $\Delta EN_{CH} = 0.4$, $\Delta EN_{HF} = 1.9$
c) $\Delta EN_{SCl} = 0.5$, $\Delta EN_{PCl} = 0.9$, $\Delta EN_{SiCl} = 1.2$

a) H—I < H—Br < H—Cl

b) H—C < H—O < H—F

c) S—Cl < P—Cl < Si—Cl

9.70 a) A solid metal is a shiny solid that conducts heat, is malleable, and melts at high temperatures. (Other answers include relatively high boiling point and good conductor of electricity.)
b) Metals lose electrons to form positive ions and metals form basic oxides.